



<Ciper/>

DevOps Roadmap

A structured, beginner-friendly path from fundamentals to production systems

13 STAGES · PREREQUISITES → ADVANCED · SELF-PACED



Roadmap Overview

PREREQUISITES → FUNDAMENTALS → CORE DEVOPS TECHNOLOGIES → ADVANCED

01
Prerequisites

02
Linux Fundamentals

03
Networking

04
Git & GitHub

05
Scripting & Automation

06
Containers

07
CI/CD

08
Cloud Fundamentals

09
Monitoring & Observability

10
Kubernetes

11
Infrastructure as Code

12
Configuration Mgmt

13 **ADVANCED**

DevSecOps · SRE · Platform Engineering · Cloud Architecture

Prerequisites

Optional groundwork before starting the roadmap. A basic grasp of programming logic and how backend systems work makes everything else easier to follow.

WHAT YOU'LL LEARN

- Programming Basics

- Backend Overview (optional)

CONCEPT FLOW

Programming Basics



Backend Concepts



Ready for DevOps

1. Linux Fundamentals

Linux runs most servers and DevOps tooling. Comfort with the command line is the foundation everything else in this roadmap builds on.

WHAT YOU'LL LEARN

- Linux Basics

- Files & Directories

- Users & Permissions

- Processes & Services

- Package Management

- Networking Commands

- Shell Basics



ESTIMATED TIME

2-3 Weeks

CONCEPT FLOW

Terminal**Shell****Processes****Filesystem**

YOUTUBE - COURSE

[liunix redhat admin 1 - Abeer Hosni](#)

2. Networking Fundamentals

DevOps engineers constantly reason about how systems talk to each other. Solid networking knowledge is essential for debugging and designing infrastructure.

WHAT YOU'LL LEARN

- OSI Model
- TCP/IP
- IP Addressing
- Subnetting
- DHCP
- HTTP / HTTPS
- Ports & Firewalls
- DNS



ESTIMATED TIME

2-3 Weeks

CONCEPT FLOW

Client



DNS



Network



Server

YOUTUBE - COURSE

[IT, Cloud, DevOps and Network](#)

3. Git & GitHub

Version control is how teams collaborate on code and infrastructure safely. Git and GitHub are used daily in almost every DevOps workflow.

WHAT YOU'LL LEARN

- Git Basics

- Branching

- Merge / Rebase

- Pull Requests

- Conflict Resolution

- GitHub Actions Basics



ESTIMATED TIME

1 Week

CONCEPT FLOW

Local Repo



Branch



Commit



Remote



Pull Request

YOUTUBE - COURSE

[Git & Github - Big Data](#)

4. Scripting & Automation

Automation is at the heart of DevOps. Shell scripting removes repetitive manual work and lets you build reliable operational tooling.

WHAT YOU'LL LEARN

- Bash
- Environment Variables
- Automation Scripts
- Cron Jobs



ESTIMATED TIME

1-2 Weeks

CONCEPT FLOW

Task**Bash Script****Cron Schedule****Automated Run**

YOUTUBE - COURSE

[Bash \(crash course\)](#)

5. Containers

Containers package applications with everything they need to run consistently. Docker is the standard tool for building and running them.

WHAT YOU'LL LEARN

- Docker Basics

- Images

- Containers

- Dockerfile

- Volumes

- Networks

- Docker Compose

- Container Registries



ESTIMATED TIME

2-3 Weeks

CONCEPT FLOW

Application**Image****Container**

YOUTUBE - COURSE

[Docker - Big Data](#)

6. CI/CD

Continuous integration and delivery automate how code moves from a commit to a running system, catching problems early and shipping faster.

WHAT YOU'LL LEARN

- CI/CD Concepts

- Artifact Management

- Pipelines

- Deployment Automation

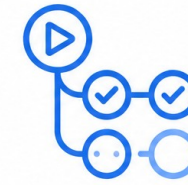
- Build

- GitHub Actions

- Test

- Jenkins / GitLab CI Basics

```
- build  
- test  
- deploy
```



CI/CD
with
GitHub Actions



ESTIMATED TIME

2 Weeks

CONCEPT FLOW

Code



Build



Test



Artifact



Deploy

YOUTUBE - COURSE

[Githu Actions \(ci/cd\) - ELfakharany](#)

7. Cloud Fundamentals

Modern infrastructure largely lives in the cloud. Understanding core cloud services helps you design, deploy, and scale systems reliably.

WHAT YOU'LL LEARN

- Cloud Concepts

- Compute

- Storage

- Networking

- IAM

- Databases

- AWS



ESTIMATED TIME

4-5 Weeks (SAA)

CONCEPT FLOW

Users



Compute



Storage & Network



IAM

YOUTUBE - COURSE

[AWS SAA - AWS Riyadh User Group](#)

8. Monitoring & Observability

You can't fix what you can't see. Monitoring and observability tools give visibility into system health, performance, and failures.

WHAT YOU'LL LEARN

- Monitoring Concepts

- Metrics

- Logs

- Alerts

- Prometheus

- Grafana

- Loki



ESTIMATED TIME

1-2 Weeks

CONCEPT FLOW

Application**Metrics & Logs****Prometheus / Loki****Grafana****SEARCH***Search: Grafana, Loki, Prometheus, exporters...*

9. Kubernetes

Kubernetes orchestrates containers at scale, handling deployment, scaling, and recovery automatically across a cluster of machines.

WHAT YOU'LL LEARN

- Kubernetes Concepts
- Pods
- Deployments
- Services
- ConfigMaps / Secrets
- Ingress
- Volumes
- Namespaces
- Helm Basics



ESTIMATED TIME

3-4 Weeks

CONCEPT FLOW

Users



Ingress



Service



Pods



Cluster

YOUTUBE - COURSE

[Kubernetes - Codographia](#)

10. Infrastructure as Code

IaC lets you define and provision infrastructure through code instead of manual steps, making environments repeatable and version-controlled.

WHAT YOU'LL LEARN

- IaC Concepts

- Terraform

- Providers

- Resources

- Variables

- State

- Modules



ESTIMATED TIME

2 Weeks

CONCEPT FLOW

Code**Terraform****Infrastructure**

YOUTUBE - COURSE

[TERRAFORM - DolphinED](#)

11. Configuration Management

Configuration management tools keep servers and environments consistent by automating setup and ongoing configuration at scale.

WHAT YOU'LL LEARN

- Configuration Mgmt Concepts

- Ansible

- Inventory

- Playbooks

- Roles



ESTIMATED TIME

1-2 Weeks

CONCEPT FLOW

Playbook**Inventory****Servers**

YOUTUBE - COURSE

[Ansible - kodeclod](#)

STAGE 12

Advanced

Once the core roadmap is complete, these areas are where DevOps careers typically specialize next. Each is a deep discipline on its own, worth exploring once the fundamentals feel solid.

01

DevSecOps

Security built into every stage of the pipeline.

02

SRE

Reliability, uptime, and incident response at scale.

03

Platform Engineering

Internal tooling that other teams build on.

04

Cloud Architecture

Designing scalable, resilient cloud systems.